An Electronic Compass Using Hall Effect Sensors

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Week 3 Report

1. Introduction

This experiment implements a compact electronic compass using two orthogonal Hall-effect sensors to resolve Earth's horizontal magnetic field into North–South (B_x) and East–West (B_y) components for heading estimation.

Because geomagnetic field changes produce only millivolt-level deviations around the 2.5 V quiescent output, an MCP6004 difference amplifier provides high-gain amplification, DC level-shifting, and low-pass filtering for stable data acquisition via the ADALM1000.

Typical applications of this kind of sensing include:

- Navigation and mobile devices — direction sensing in smartphones, tablets and handheld GPS receivers.
- Robotics and autonomous systems — heading estimation for drones, ground robots and SLAM-based navigation.
- Automotive systems — digital compasses, dashboard orientation tracking and driver-assist modules.
- Industrial and geophysical tools — magnetic surveying, alignment systems and field instruments.
- Marine and aviation — heading reference independent of vehicle motion.

2. Components and Tools Used

- Hall Effect Sensor — 2, mounted orthogonally
- Operational Amplifier — MCP6004 (quad op-amp package)
- Resistors — 500 Ω (input/gain-setting) and 695 kΩ (feedback)
- Capacitor — 10 nF, placed in parallel with the feedback resistor for band-limiting / noise filtering
- DC Supply and Measurement Unit — ADALM1000 (5 V DC supply, used in Split I/O mode for biasing and acquisition)
- Solderless Breadboard and connecting wires
- ALICE software — for waveform acquisition and averaging
- A smartphone digital compass — used as the ground-truth reference direction while rotating the sensor assembly

3. Circuit Design and Calculations

3.1 Gain and Offset Requirement

The sensitivity of the Hall sensor is 1.8 mV per 100 µT. The op-amp is powered from a single 5 V supply with the non-inverting input biased at the mid-supply common-mode voltage of 2.5 V, which converts the small bipolar Hall swing into a usable unipolar output.

For the op-amp output to saturate at a nominal field of 300 µT, an output swing of 2.5 V is required for an input change of 1.8 × 3 = 5.4 mV, giving:

Gain = 2500 / 5.4 ≈ 463

so that the required difference-amplifier gain equation is 1 + R2/R1 ≈ 463. For the simulation and hardware build used in this report, R2 = 695 kΩ and R1 = 500 Ω were chosen instead, which raises the gain and lets the amplifier reach saturation at a smaller field of about 100 µT — this was done deliberately to make the saturation behaviour easy to observe within the sweep range used for testing.

3.2 Simulation Setup

The Hall sensor was modelled in LTspice as a DC voltage source producing an output centred at VDD/2 (2.5 V for a 5 V supply). A .step param directive was used to sweep the modelled Hall voltage VHal from 2.4982 V to 2.5018 V in steps of 0.18 mV, i.e. ±1.8 mV about the 2.5 V centre, corresponding to a swept magnetic field of −100 µT to +100 µT given the sensor's 1.8 mV/100 µT sensitivity.

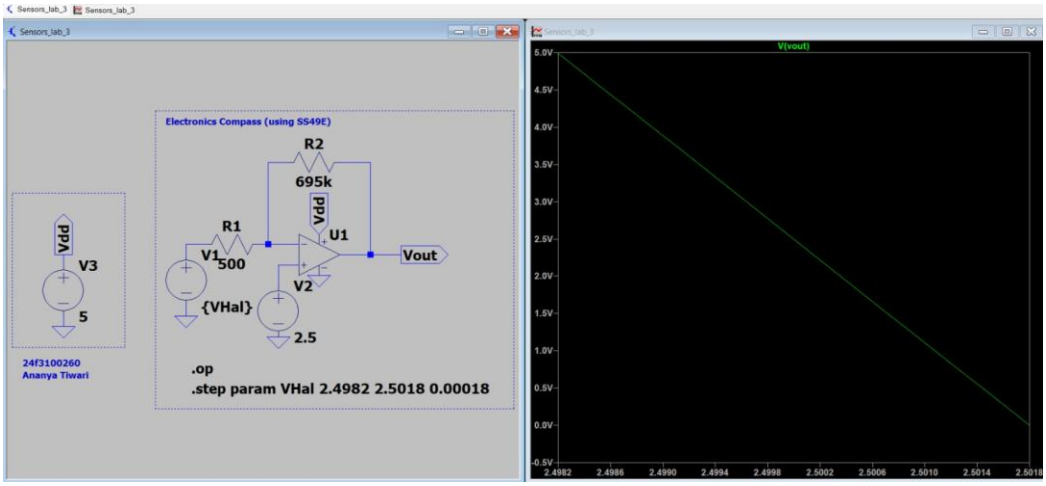


Figure 1: Simulated circuit of the Hall sensor with difference amplifier (LTspice) and the resulting amplifier output.

3.3 Simulation Results

The simulated output Vout falls almost perfectly linearly from about 5 V down to 0 V as VHal is swept from 2.4982 V to 2.5018 V, confirming that the inverting difference-amplifier configuration converts the small bipolar Hall excursion into a large, nearly rail-to-rail output swing centred at mid-supply. This linear relationship is the basis on which the heading angle can later be reconstructed from the amplified X and Y voltages.

4. Hardware Implementation

4.1 Breadboard Setup

The signal-conditioning circuit was built on a solderless breadboard around the MCP6004 quad op-amp. The Hall sensor output is coupled through the 500 Ω input resistor to the inverting terminal, the 695 kΩ feedback resistor sets the gain, and the 10 nF capacitor is placed across the feedback path to band-limit the amplifier and suppress high-frequency noise. Red jumpers carry the +5 V and ground rails, while the coloured flying leads bring the Hall sensor output and the ADALM1000 bias/measurement lines to the correct breadboard rows.

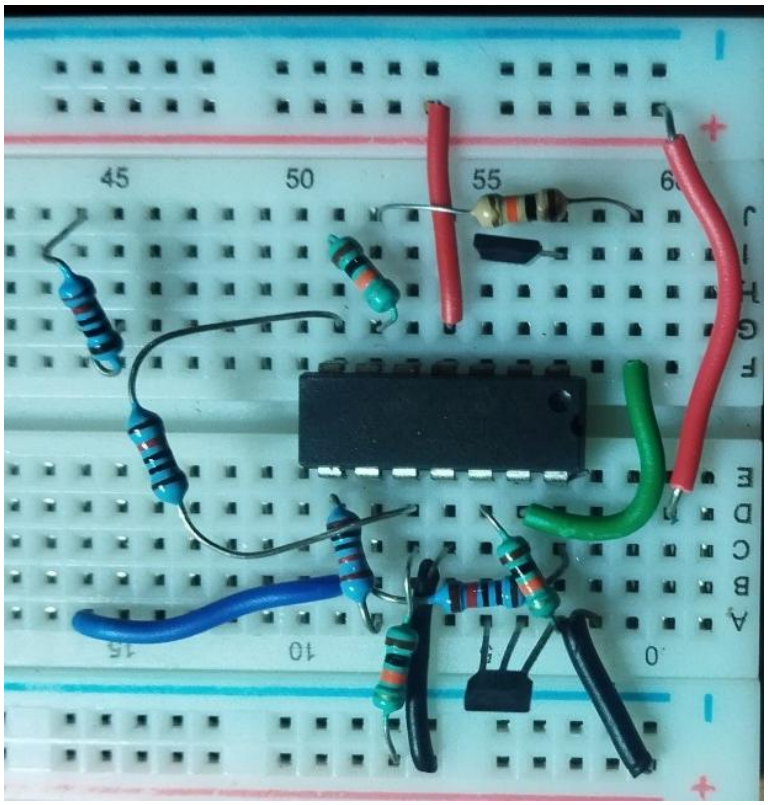


Figure 6: Breadboard hardware implementation of the Hall-sensor difference amplifier.

4.2 Mounting of Hall ICs

Two Hall sensors were mounted at 90° to each other on a non-magnetic cardboard strip, with their sensitive axes aligned to the orthogonal horizontal components of the Earth's magnetic field. This orientation allows the X-axis sensor to primarily respond to the North–South component and the Y-axis sensor to the East–West component of the local field as the assembly is rotated.

4.3 Circuit Wiring

The Hall sensor's output is connected to the inverting terminal of the op-amp through the 500 Ω resistor. The non-inverting terminal receives a precise 2.5 V DC offset generated using Channels A and B of the ADALM1000 rather than a trim-pot, since the high gain of the amplifier makes trim-pot tuning prone to error; the ADALM1000 offers finer resolution and better stability for this biasing. The feedback loop includes the gain resistor in parallel with the 10 nF capacitor for low-pass filtering.

4.4 Measurement Procedure and Noise Filtration

Measurements were carried out using the ALICE / ADALM1000 oscilloscope software with the device operated in Split I/O mode, which allows simultaneous voltage sourcing and measurement on the same channel — useful here since two independent bias voltages are needed for the difference amplifier inputs. The 10 nF feedback capacitor was used throughout to filter high-frequency noise so that the steady-state channel averages could be read reliably.

5. Testing and Data Recording

5.1 Data Collection Method

The mounted sensor assembly was rotated by hand and checked against a smartphone digital compass used as the ground-truth reference. At each of the four inter-cardinal headings — North-East (NE), North-West (NW), South-

West (SW) and South-East (SE) — the amplifier outputs were connected to Channel A (VoutX) and Channel B (VoutY) of the ADALM1000, and the steady-state channel averages were recorded once the trace had settled.

5.2 Readings from ADALM1000

Direction	Ch-A, VoutX (V)	Ch-B, VoutY (V)
North-East (NE)	3.133	1.054
North-West (NW)	0.384	4.844
South-West (SW)	0.199	0.199
South-East (SE)	4.743	0.336

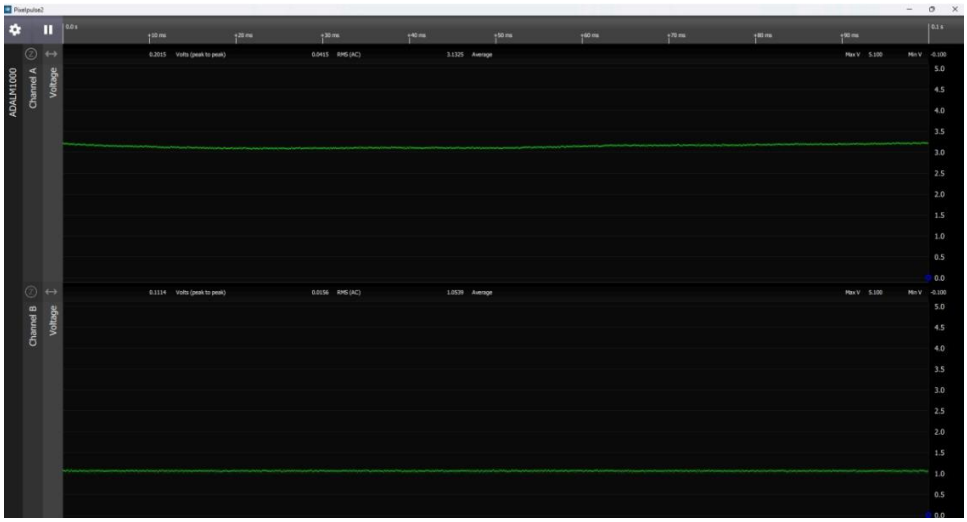


Figure 2: Facing North-East — Channel A average 3.1325 V, Channel B average 1.0539 V.

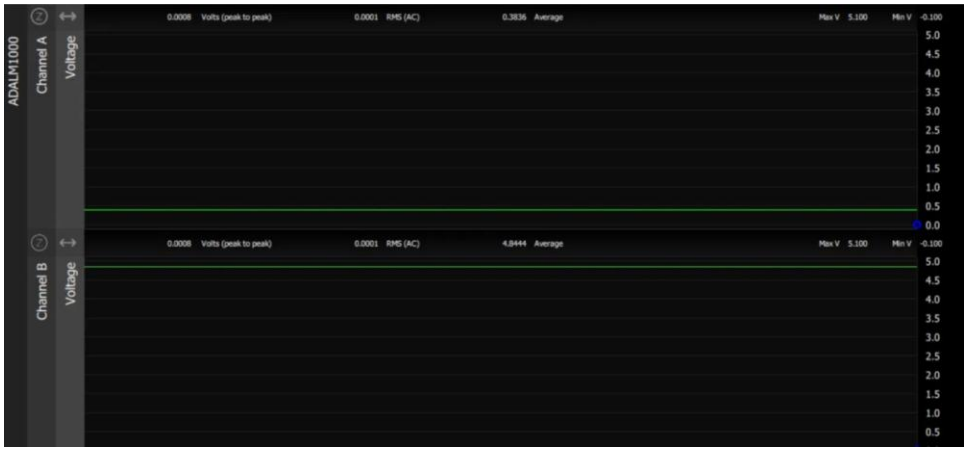


Figure 3: Facing North-West — Channel A average 0.3836 V, Channel B average 4.8444 V.

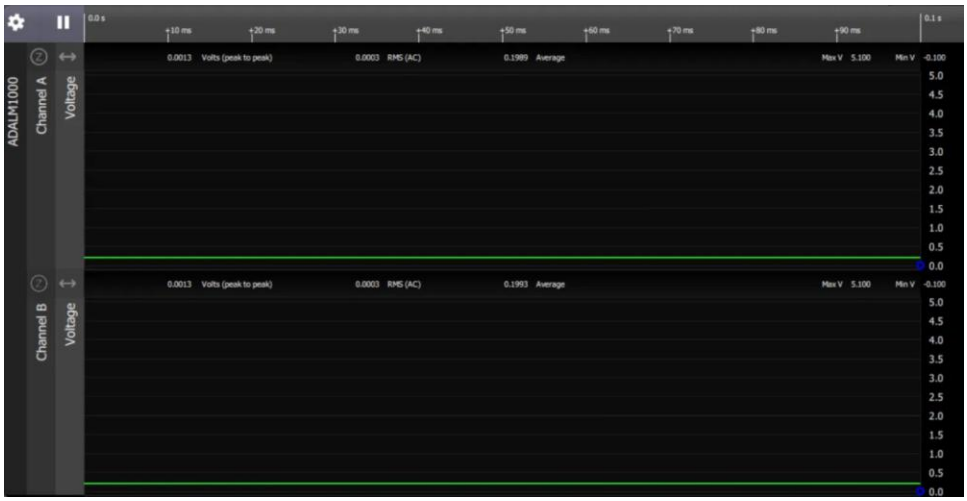


Figure 4: Facing South-West — Channel A average 0.1989 V, Channel B average 0.1993 V.



Figure 5: Facing South-East — Channel A average 4.7434 V, Channel B average 0.3363 V.

6. Data and Analysis

Plotting V_{outY} against V_{outX} traces the horizontal magnetic field vector across four headings: NW lies along the Y-axis (high $V_{outY} \approx 4.84$ V, low $V_{outX} \approx 0.38$ V), SW shows both channels low (≈ 0.20 V), and SE shifts primarily to the X-axis (high $V_{outX} \approx 4.74$ V, low $V_{outY} \approx 0.34$ V).

The off-corner NE reading ($V_{outX} \approx 3.13$ V, $V_{outY} \approx 1.05$ V) indicates channel gain or offset mismatch rather than a failure in the sensing principle.

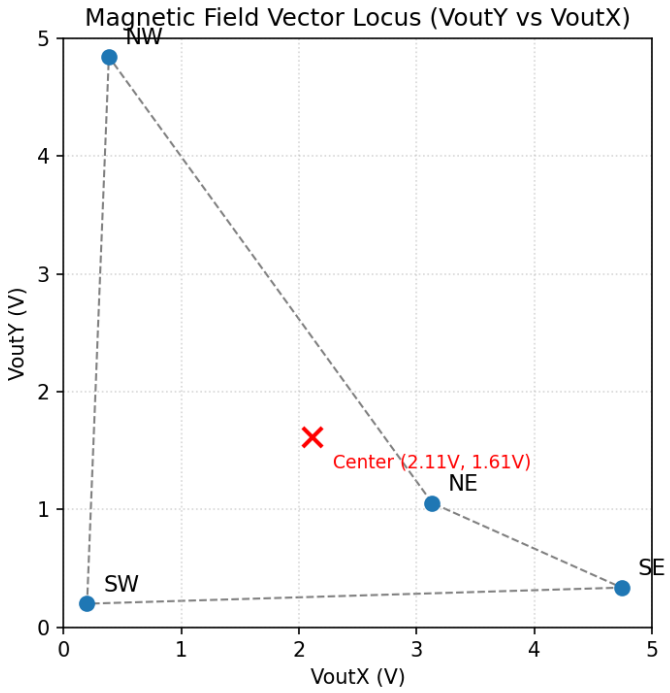


Figure 7: Magnetic field vector locus (VoutY vs VoutX) for the four measured headings.

The computed centroid of the four points (≈ 2.11 V, 1.61 V) is offset from the ideal mid-supply centre of (2.5 V, 2.5 V), and the quadrilateral joining the points is noticeably skewed rather than a symmetric square. This is consistent with a combination of offset error introduced by the ADALM1000 bias source, small gain mismatch between the two amplifier channels, and imperfect mechanical alignment of the two Hall sensors on the cardboard strip.

7. Discussion

7.1 Performance of the Experimental Setup

The electronic compass successfully validated orthogonal geomagnetic sensing, producing distinct, wide dynamic range output swings across rotation.

However, channel gain/offset mismatches, sensor misalignment, and mechanical tilt skewed the locus and shifted the NE point away from the expected high-high corner. Consequently, computing heading via $\arctan2(V_{outY}, V_{outX})$ incurs several degrees of error prior to calibration.

7.2 Possible Improvements

- Calibration and gain matching — record VoutX and VoutY over a full 360° rotation, fit an ellipse to the locus, and use the resulting offset and scale factors in software to re-centre the data before computing the heading.
- Match the two amplifier gains more closely using precision resistors or a digital gain-trim, which directly reduces the heading error produced by the atan2 calculation.
- Use a 3-axis digital magnetometer (e.g. HMC5883L) for improved accuracy and easier digital calibration.
- Implement digital filtering (e.g. a moving average) on the acquired $\frac{VoutX}{VoutY}$ samples to further suppress noise.
- Stream the ADALM1000 data in real time so the heading can be computed and displayed continuously rather than from four static readings.

8. Conclusion

An electronic compass was implemented using two orthogonal Hall sensors conditioned by an MCP6004 difference amplifier. Hardware tests confirmed that the circuit successfully amplified the millivolt-level sensor outputs into near rail-to-rail swings across four headings.

The resulting V_{outY} vs. V_{outX} locus showed expected qualitative rotation but exhibited real-world skew caused by channel gain mismatch, sensor misalignment, and bias offsets—errors typically corrected via calibration or ellipse fitting. Overall, the experiment bridged LTspice simulations with practical hardware challenges using the ADALM1000/ALICE toolchain.